

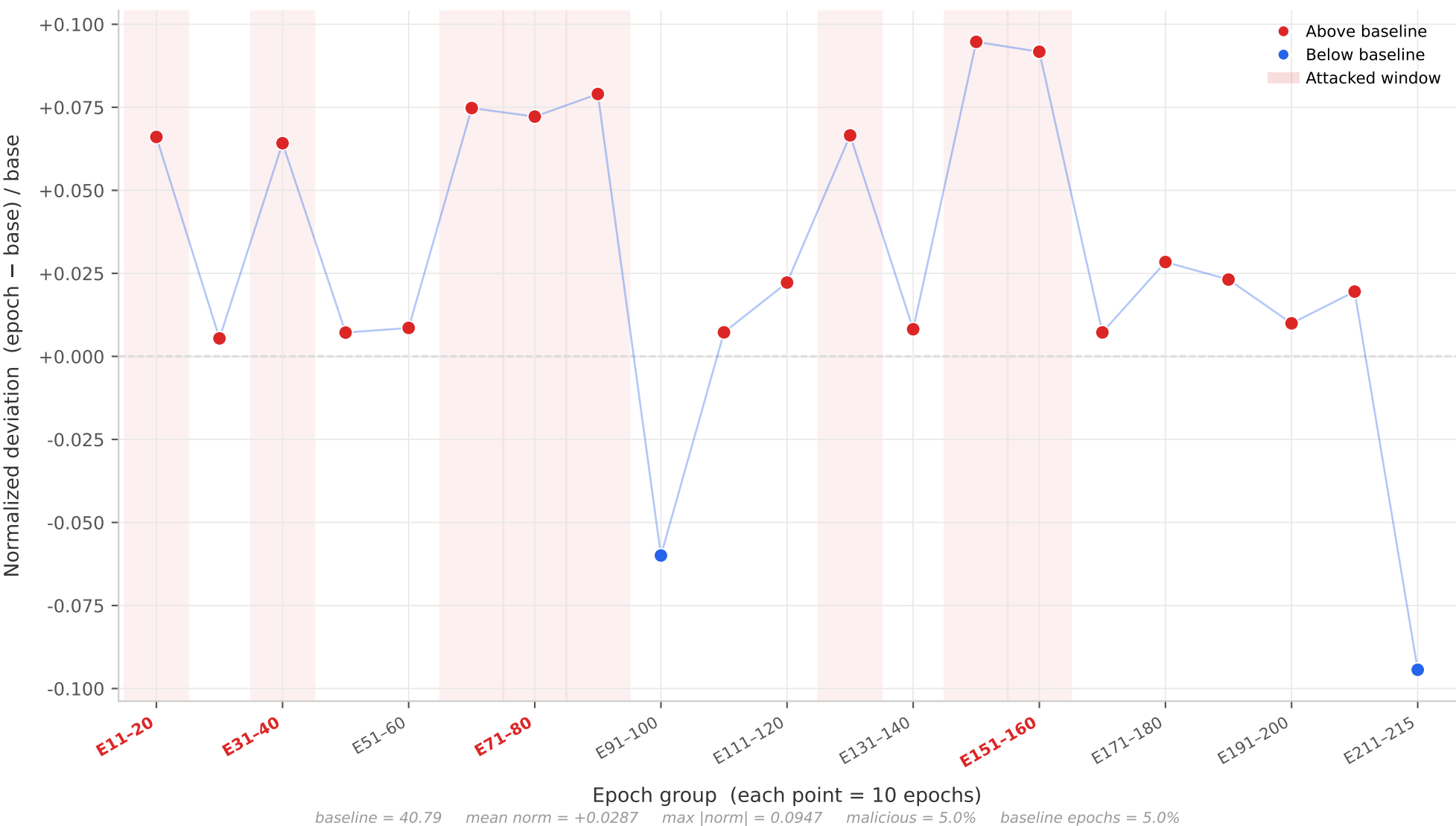
Entropy Stream Analysis

Deviation from Baseline — Per-Metric Report

Run Parameters		Baseline Values	
Dataset	epochs_5.0_5.0_sparse.json	Src IP Entropy	40.79
Directory	CAIDA_Dataset/txtD280/epochs/	Dst IP Entropy	32.08
Timestamp	2026-02-23T21:03:00.970882	Src Port Entropy	22.05
Total epochs	215	Dst Port Entropy	43.05
Baseline epochs	10 (5.0%)	Src+Dst IP Entropy	52.55
Attack epochs	205	Src+Dst Port Entropy	53.96
Malicious traffic	5.0%	Src IP Cardinality	5022
Epoch group size	10 epochs per point	Dst IP Cardinality	5005
Metrics plotted	20	Src Port Cardinality	4288
Mode	SPARSE	Dst Port Cardinality	5020
Attack probability	0.3	Src+Dst IP Cardinality	5040
Windows total	21	Src+Dst Port Cardinality	5037
Windows attacked	8	Src IP Uniformity	3.919e+08
Windows clean	13	Dst IP Uniformity	1.085e+06
		Src/Dst Ratio	1.004
		Total Flows	5056

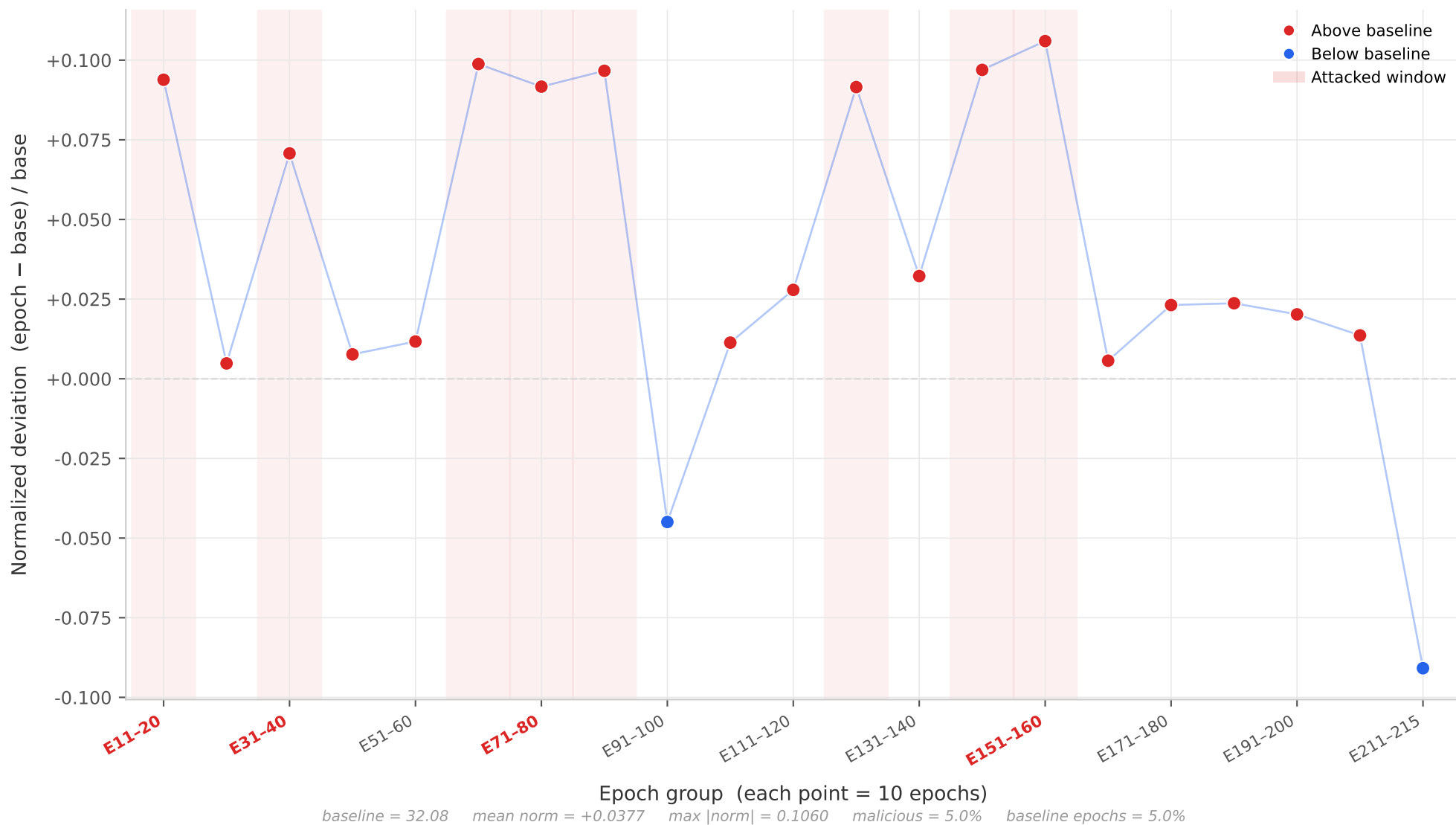
Source IP Entropy

Entropy



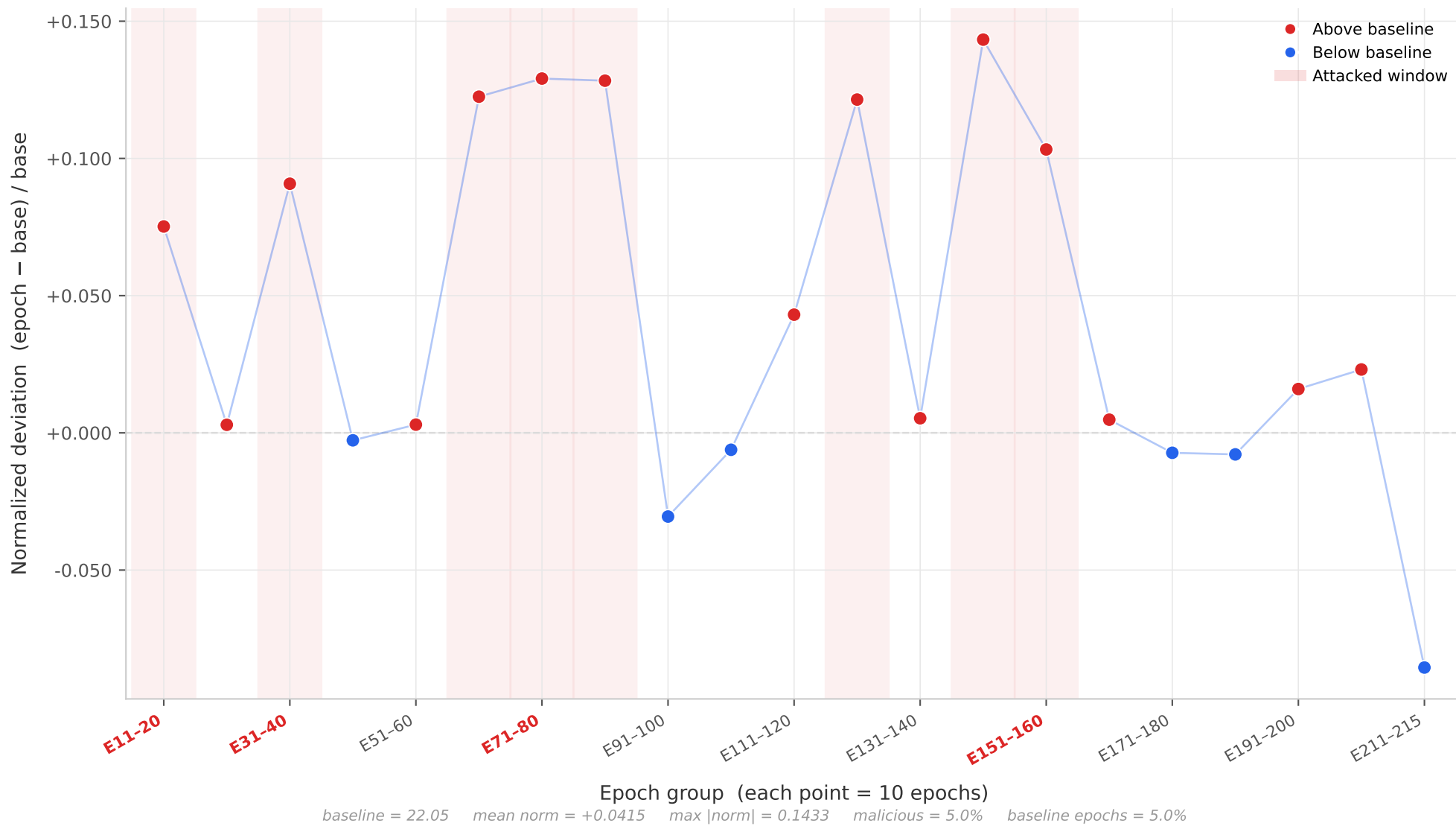
Destination IP Entropy

Entropy



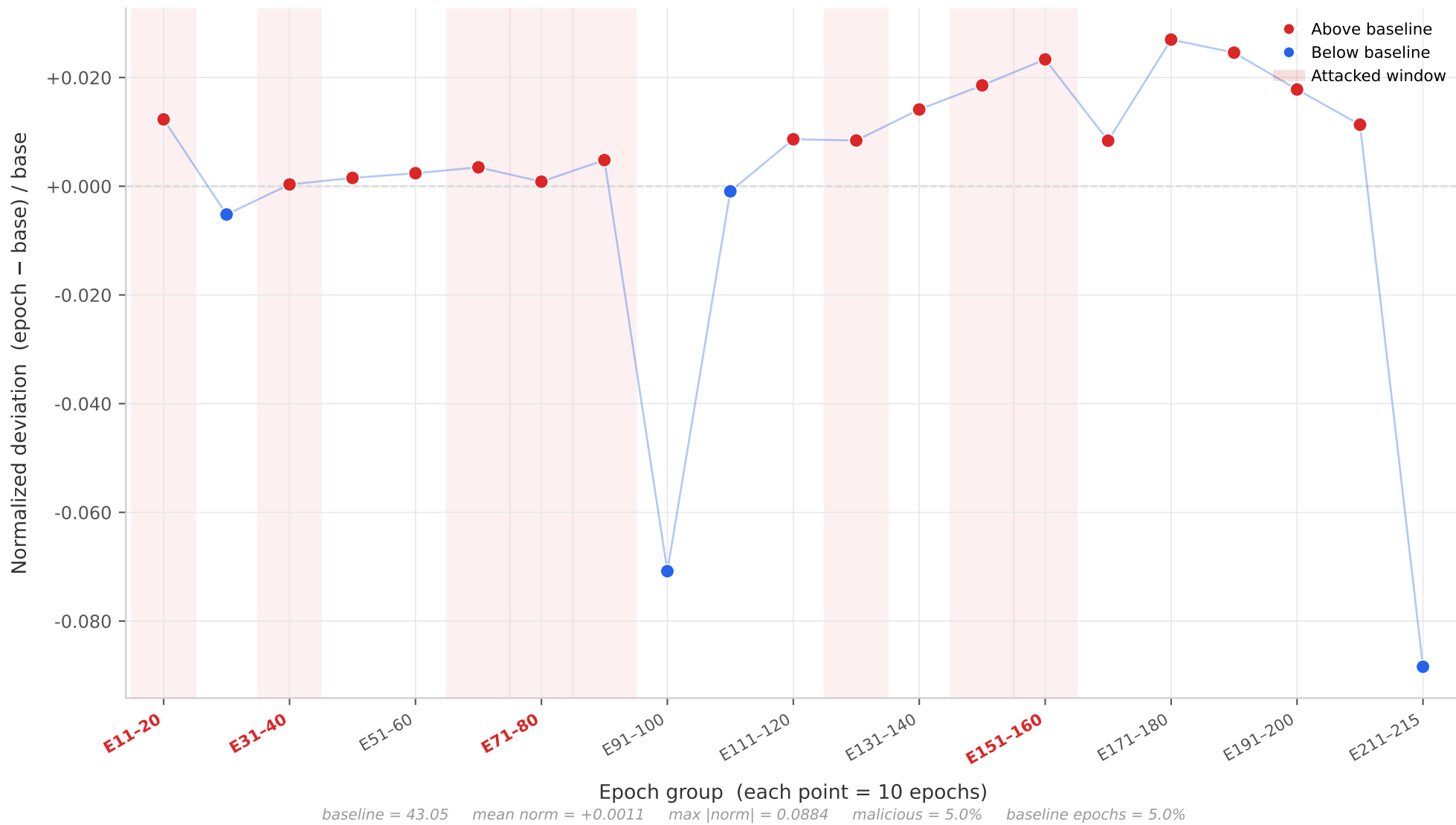
Source Port Entropy

Entropy



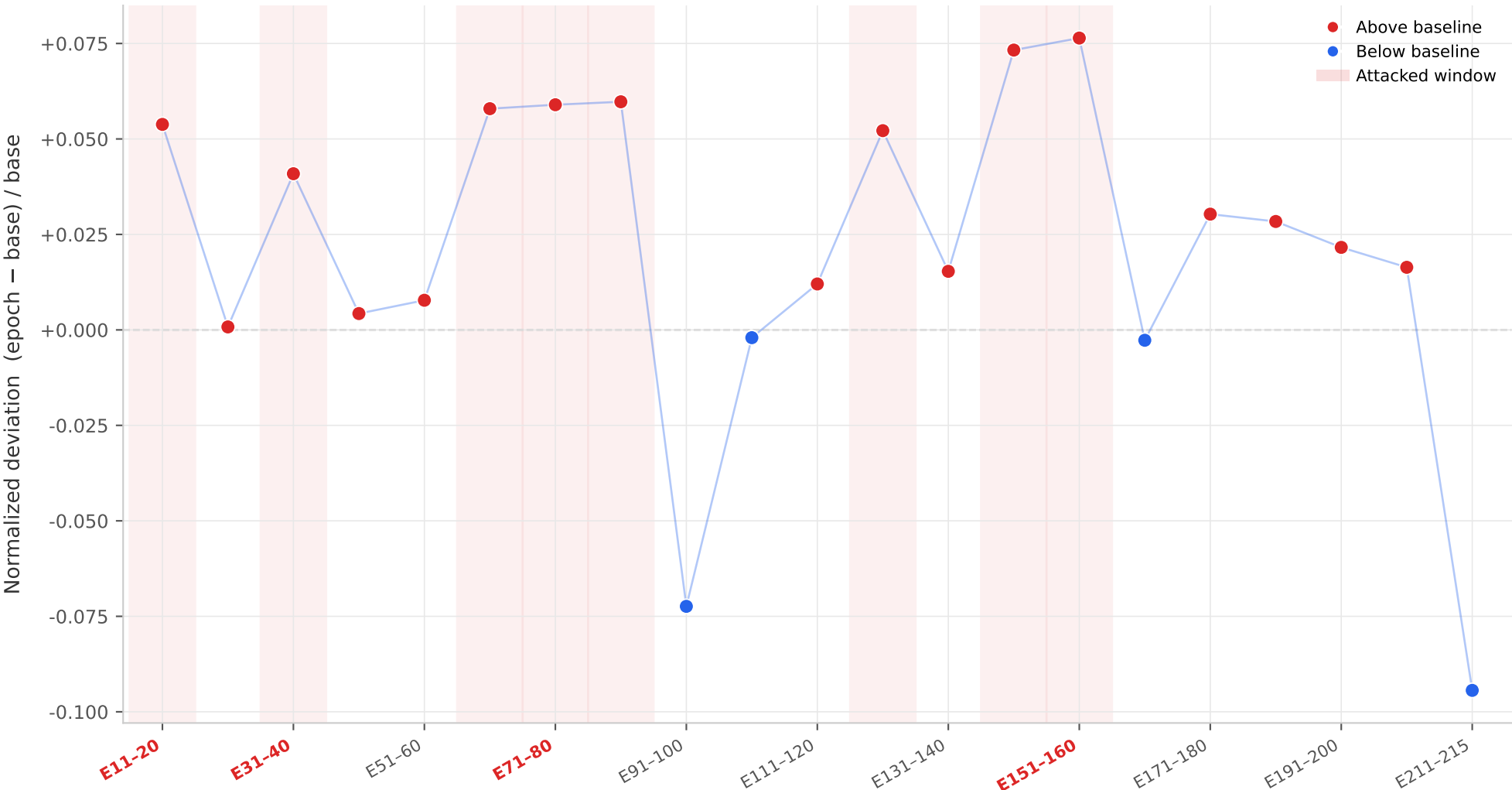
Destination Port Entropy

Entropy



Src+Dst IP Pair Entropy

Entropy

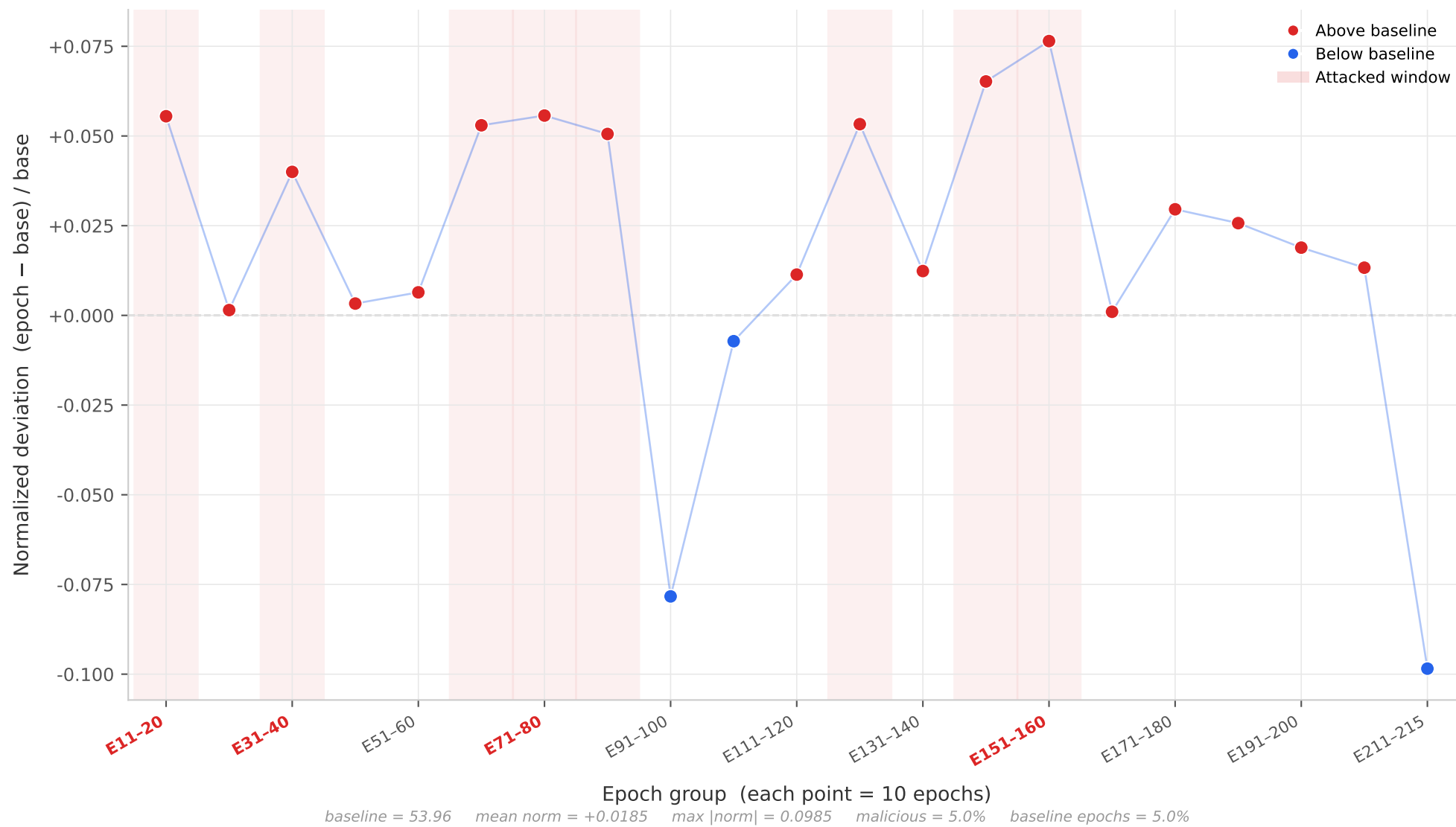


Epoch group (each point = 10 epochs)

baseline = 52.55 mean norm = +0.0209 max |norm| = 0.0944 malicious = 5.0% baseline epochs = 5.0%

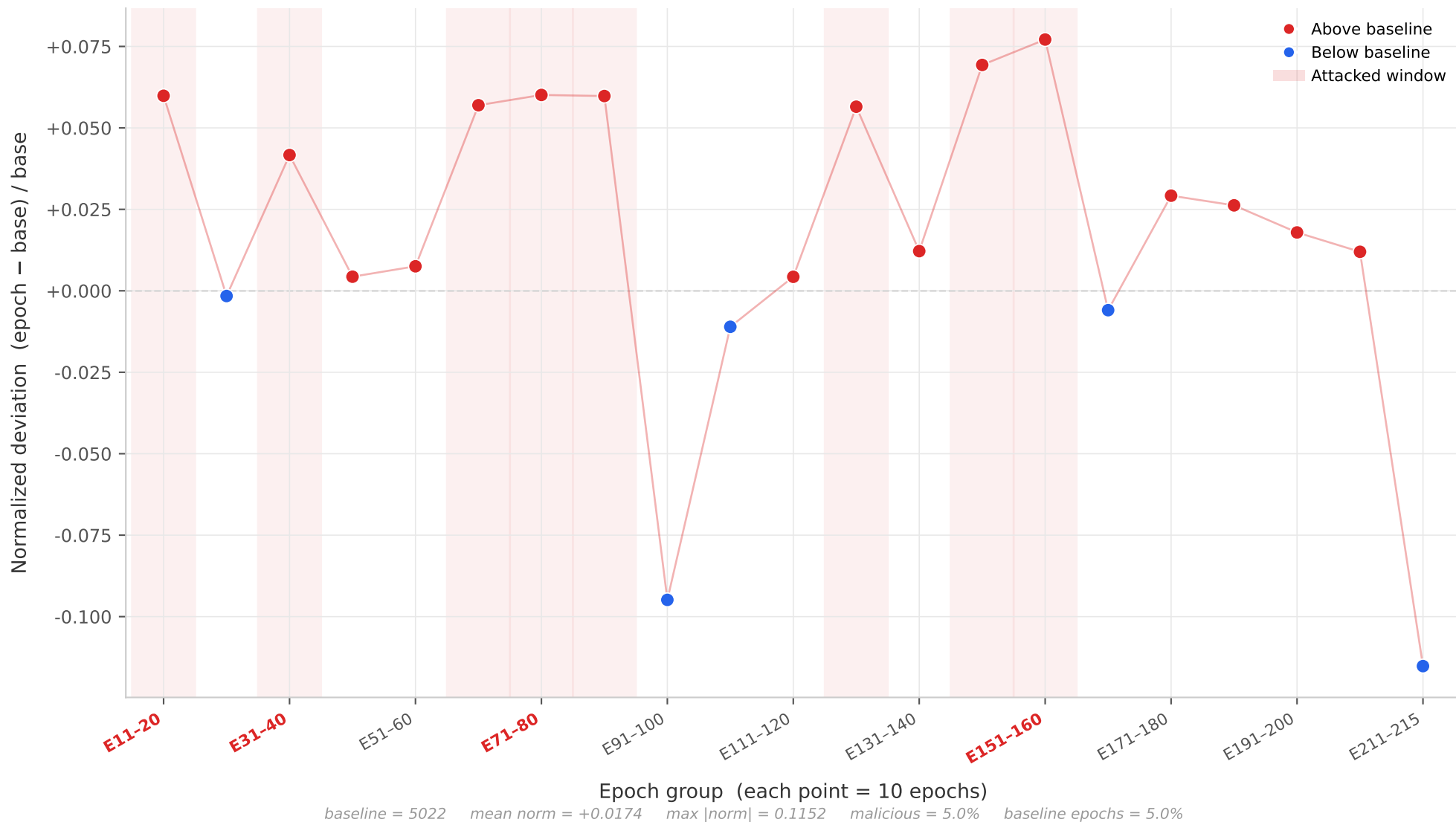
Src+Dst Port Pair Entropy

Entropy



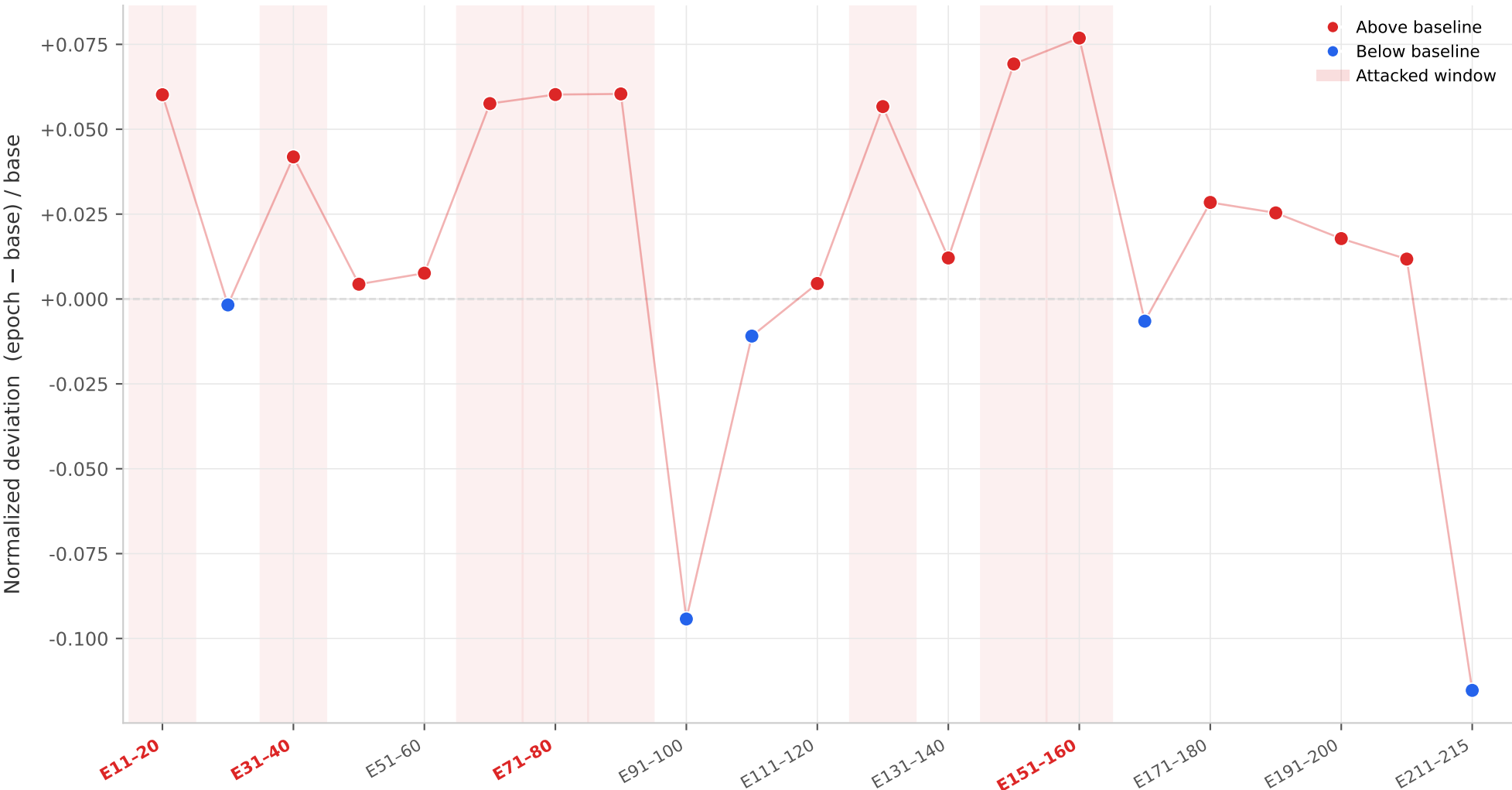
Source IP Cardinality

Cardinality



Destination IP Cardinality

Cardinality

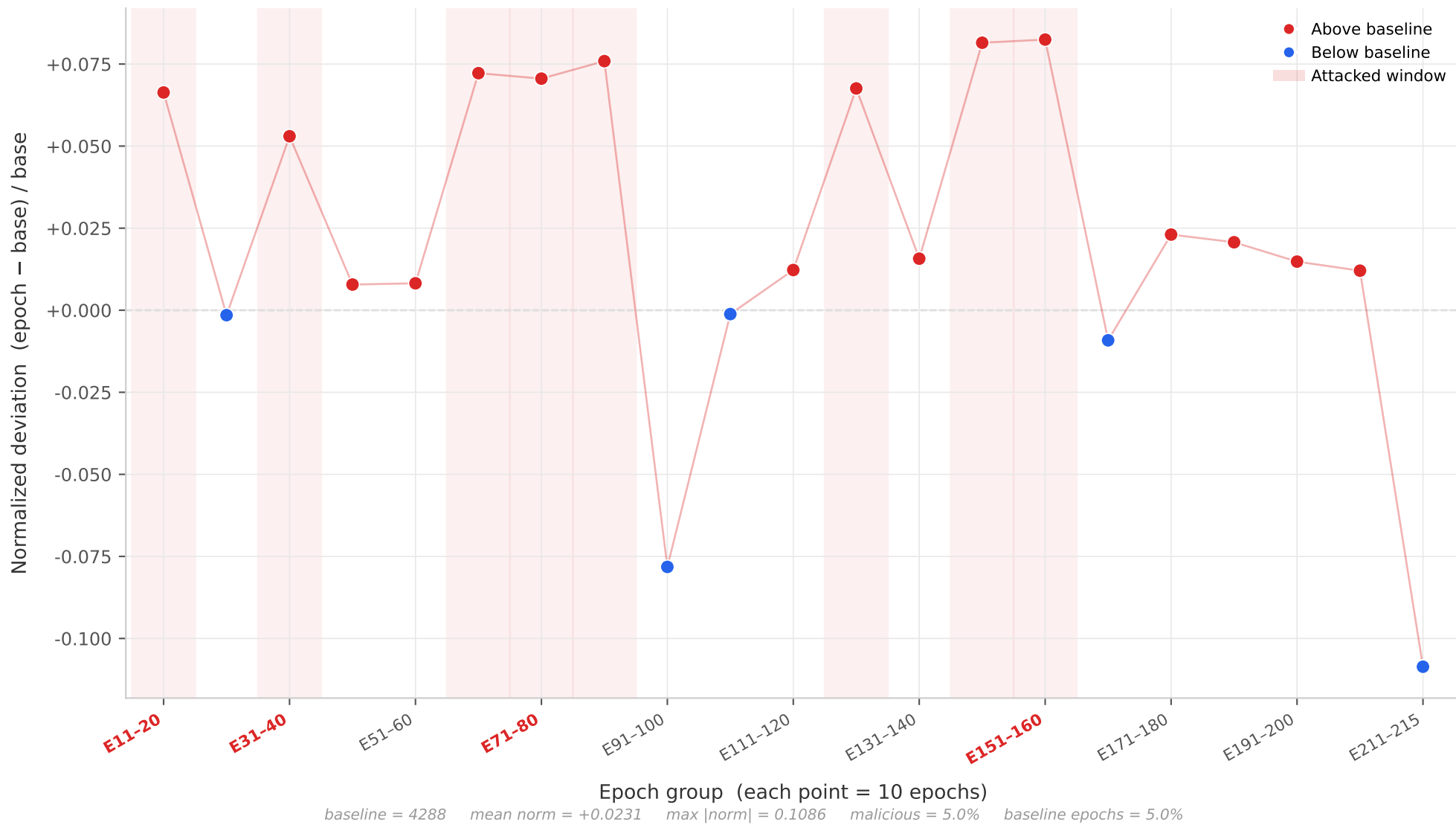


Epoch group (each point = 10 epochs)

baseline = 5005 mean norm = +0.0174 max |norm| = 0.1153 malicious = 5.0% baseline epochs = 5.0%

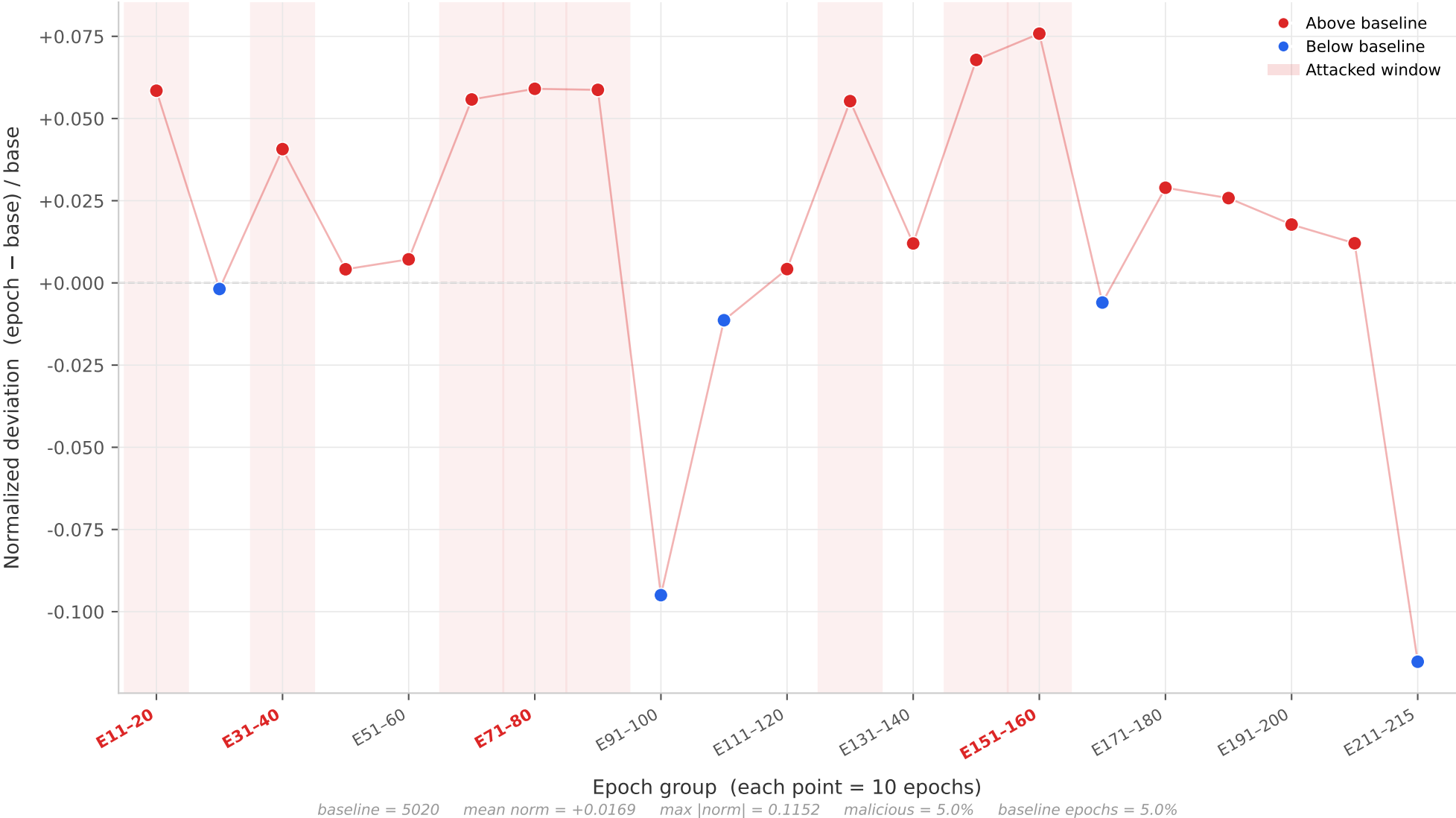
Source Port Cardinality

Cardinality



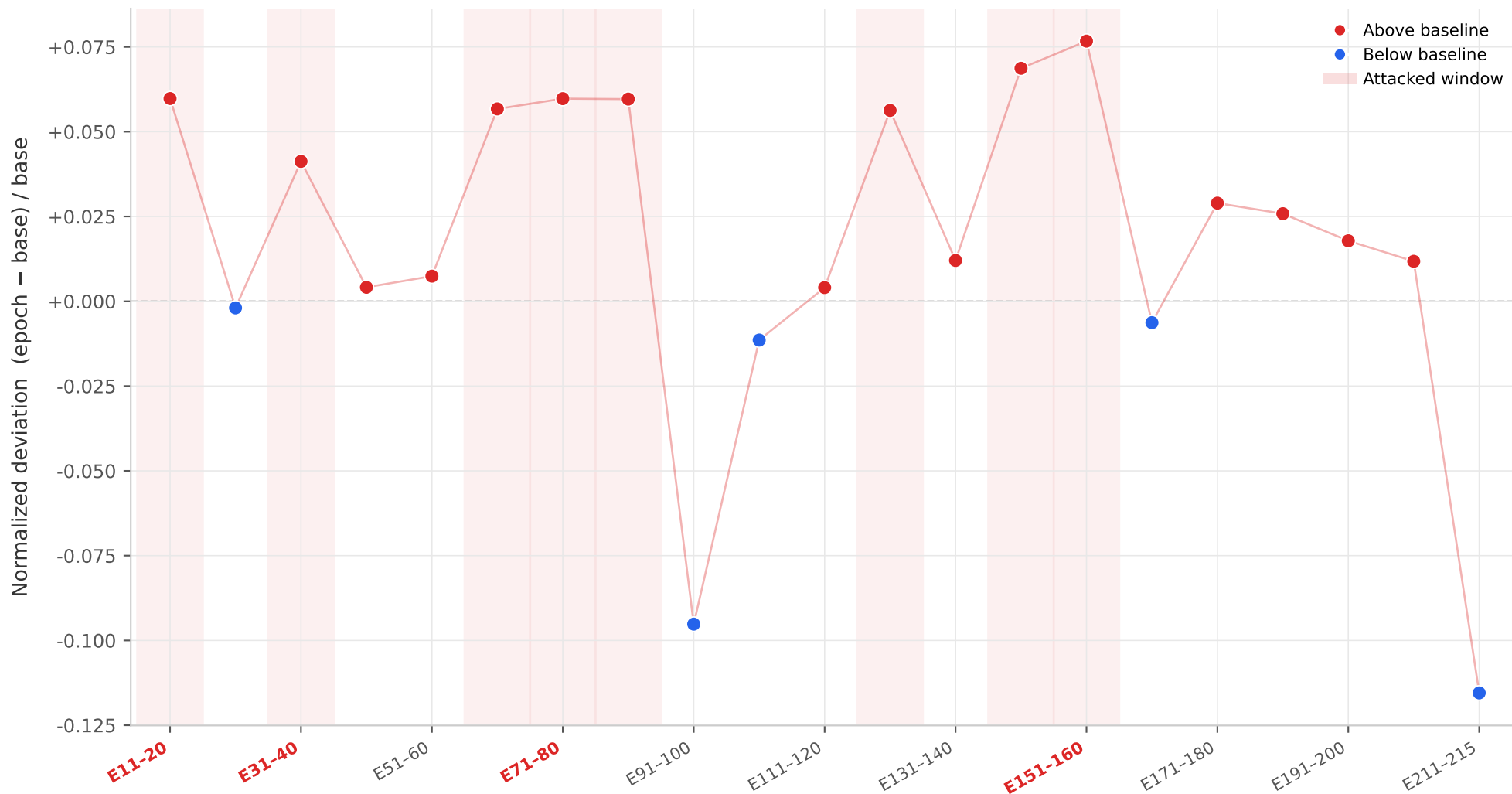
Destination Port Cardinality

Cardinality



Src+Dst IP Pair Cardinality

Cardinality

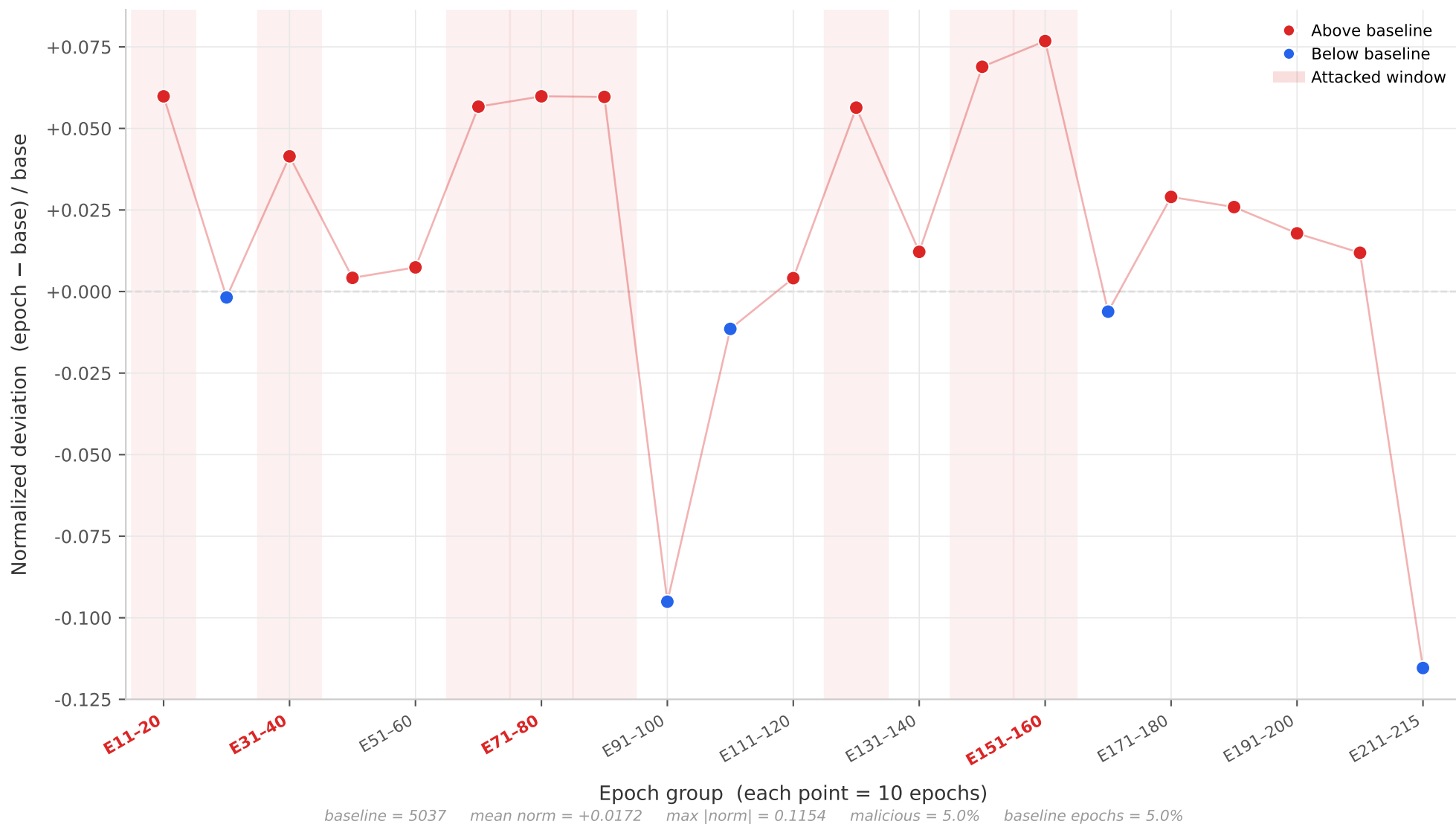


Epoch group (each point = 10 epochs)

baseline = 5040 mean norm = +0.0172 max |norm| = 0.1155 malicious = 5.0% baseline epochs = 5.0%

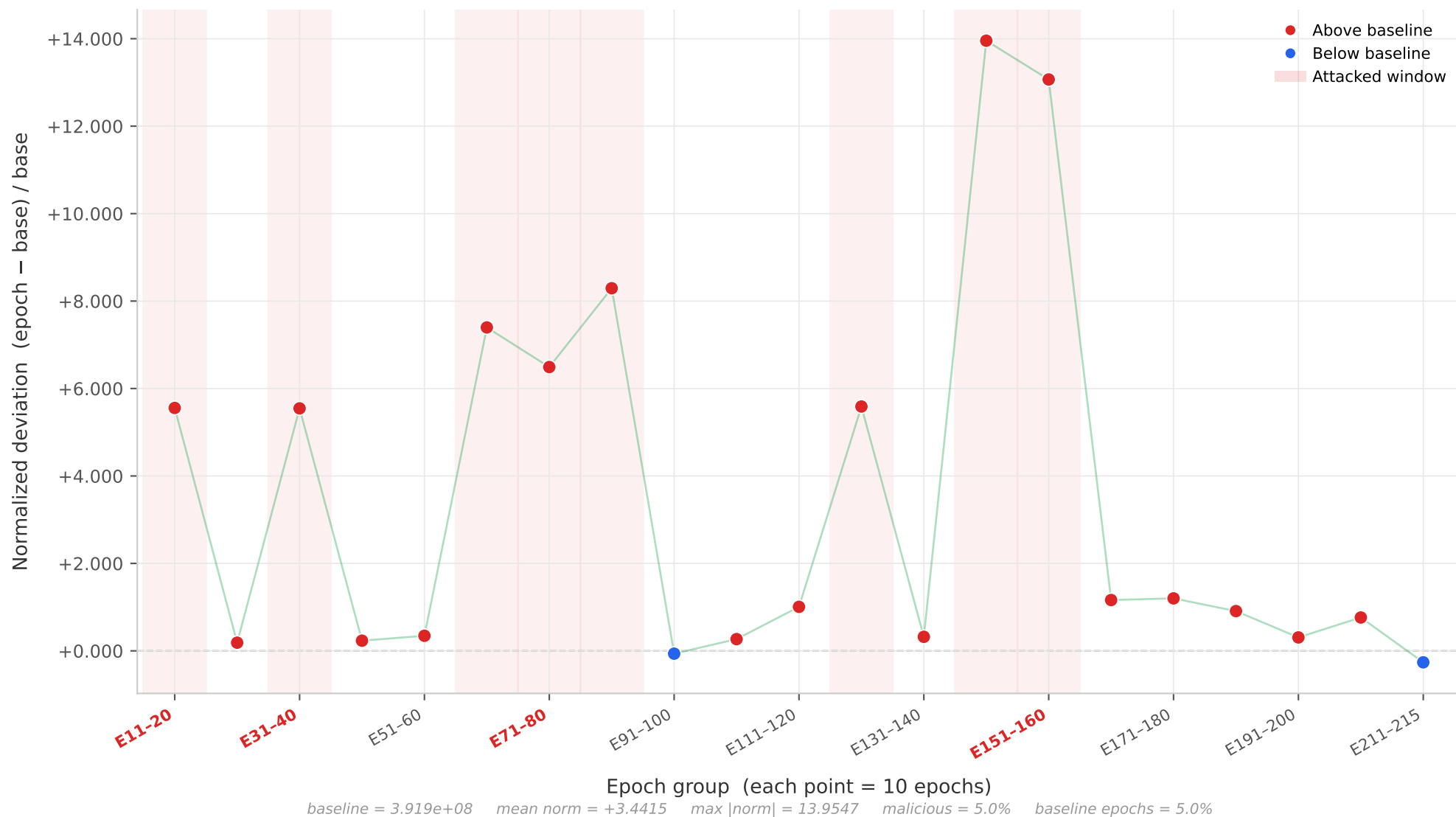
Src+Dst Port Pair Cardinality

Cardinality



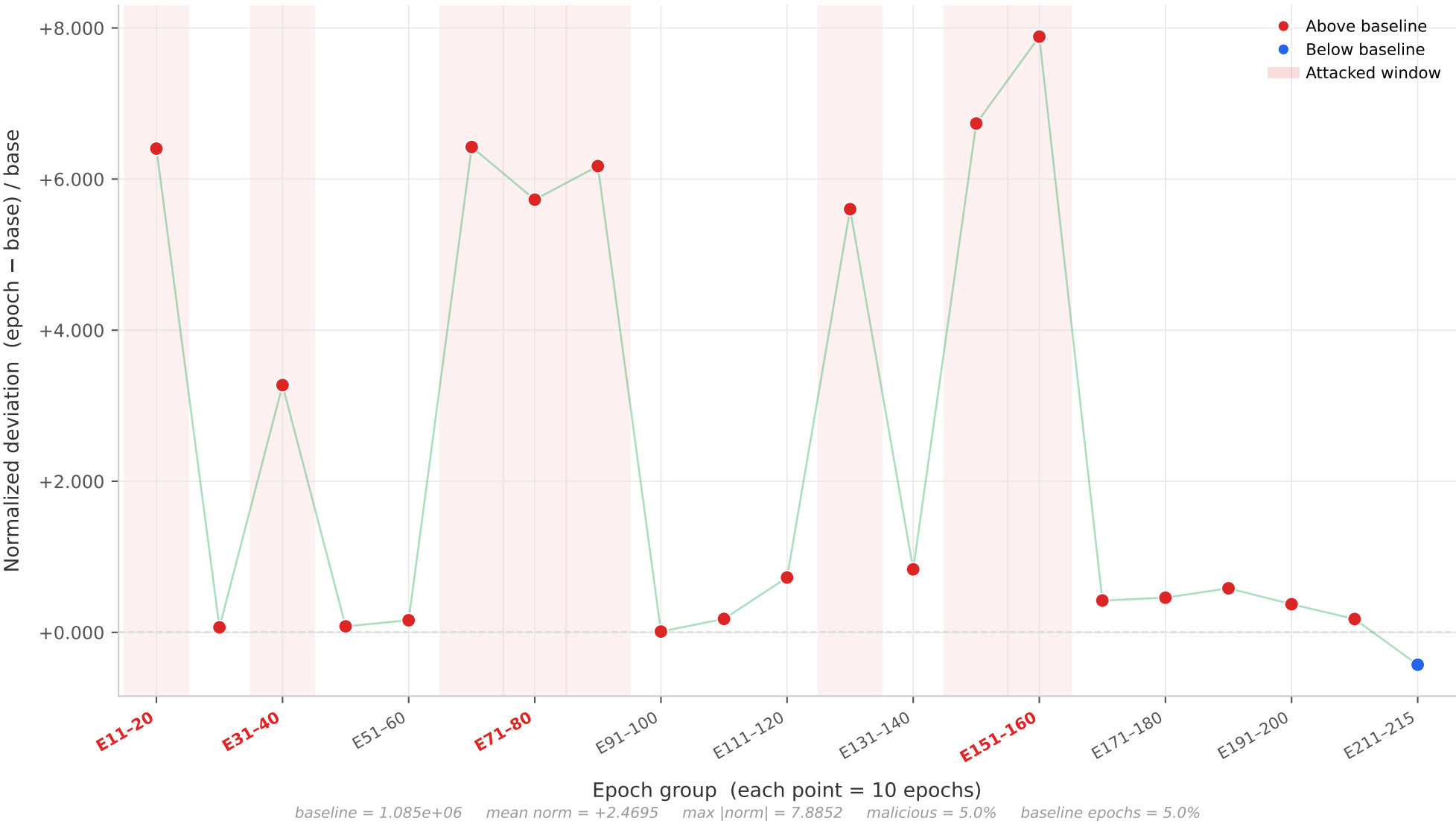
Source IP Uniformity

Uniformity



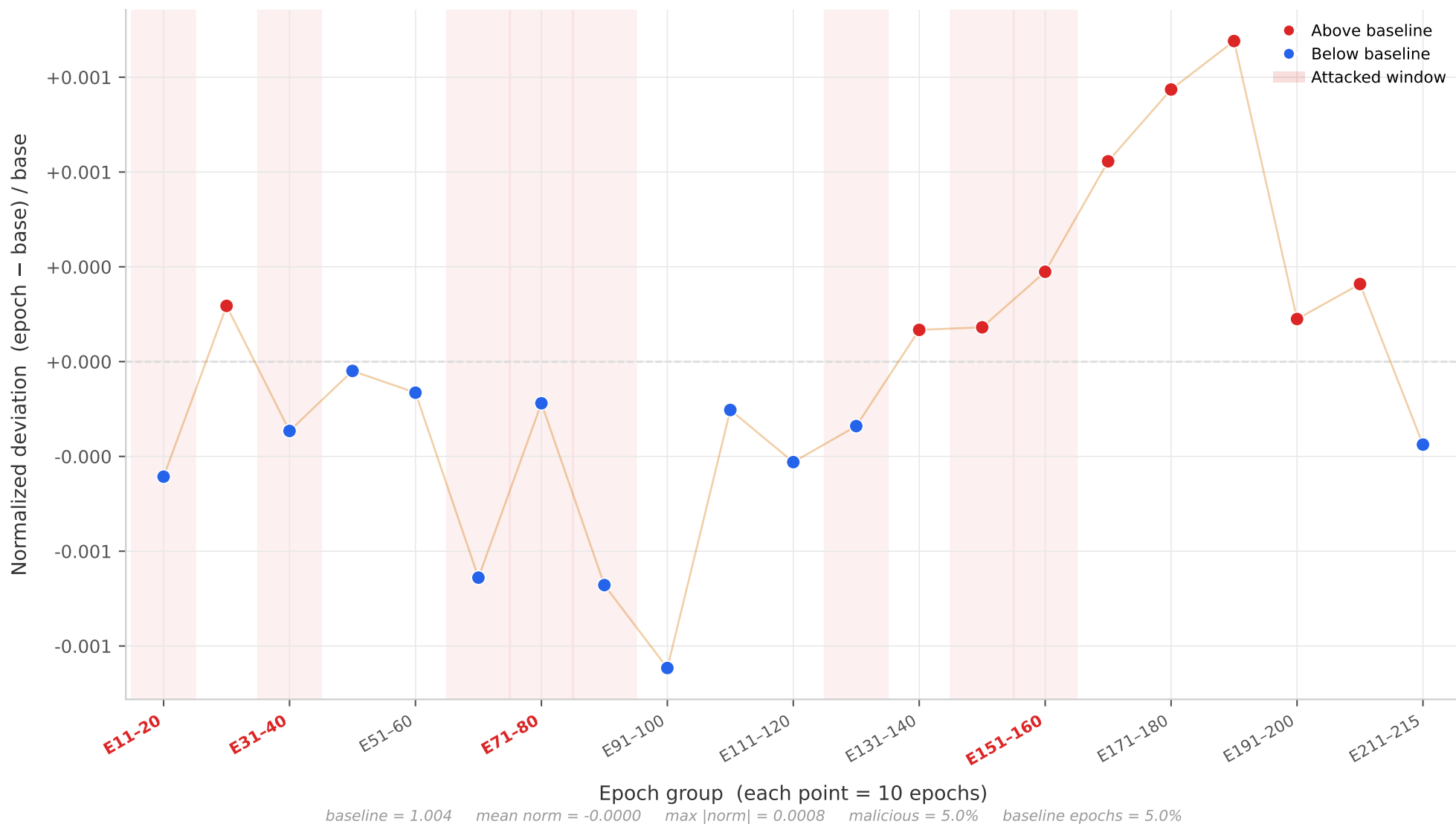
Destination IP Uniformity

Uniformity



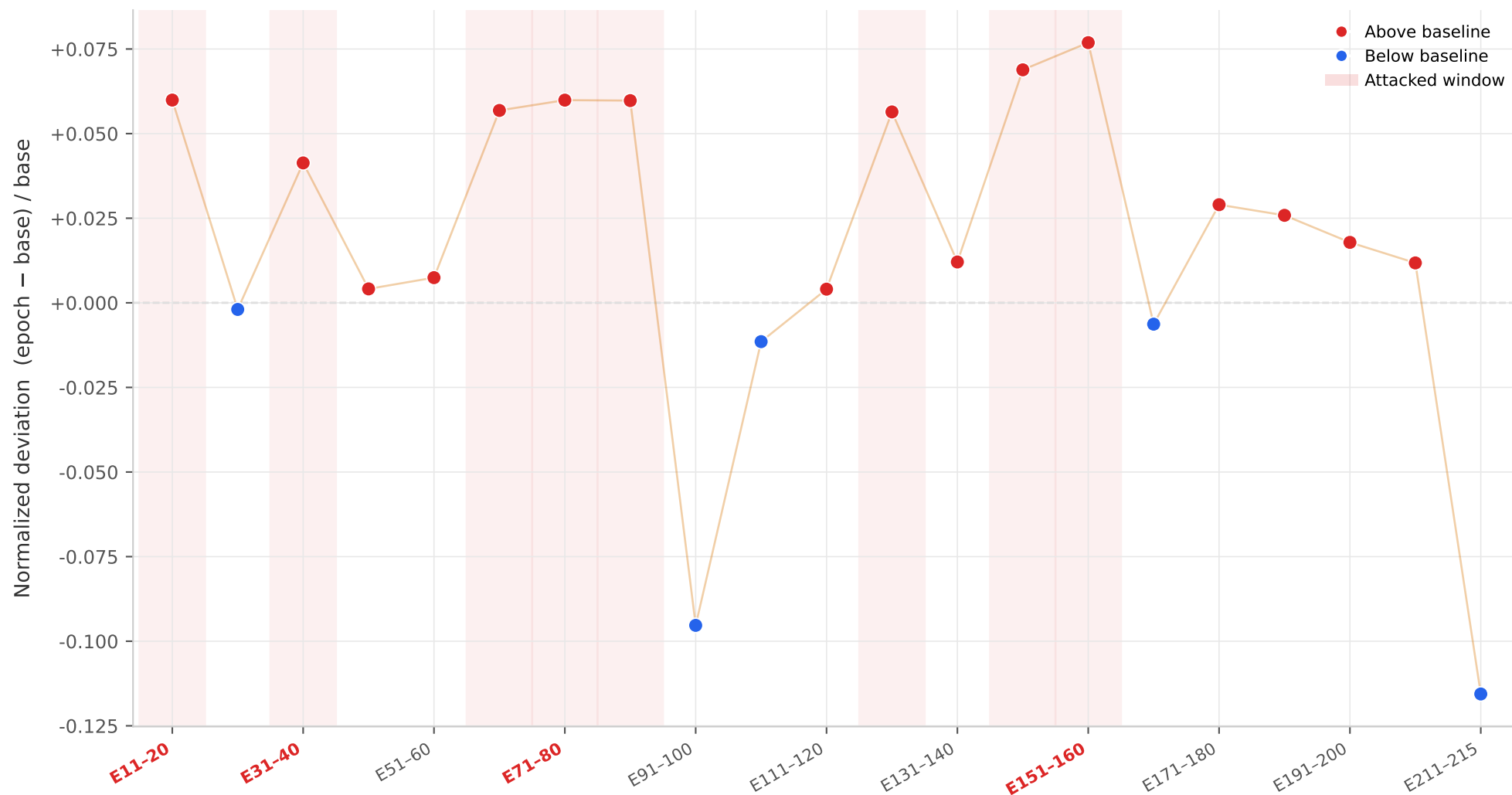
Src / Dst IP Ratio

Flow



Total Flows

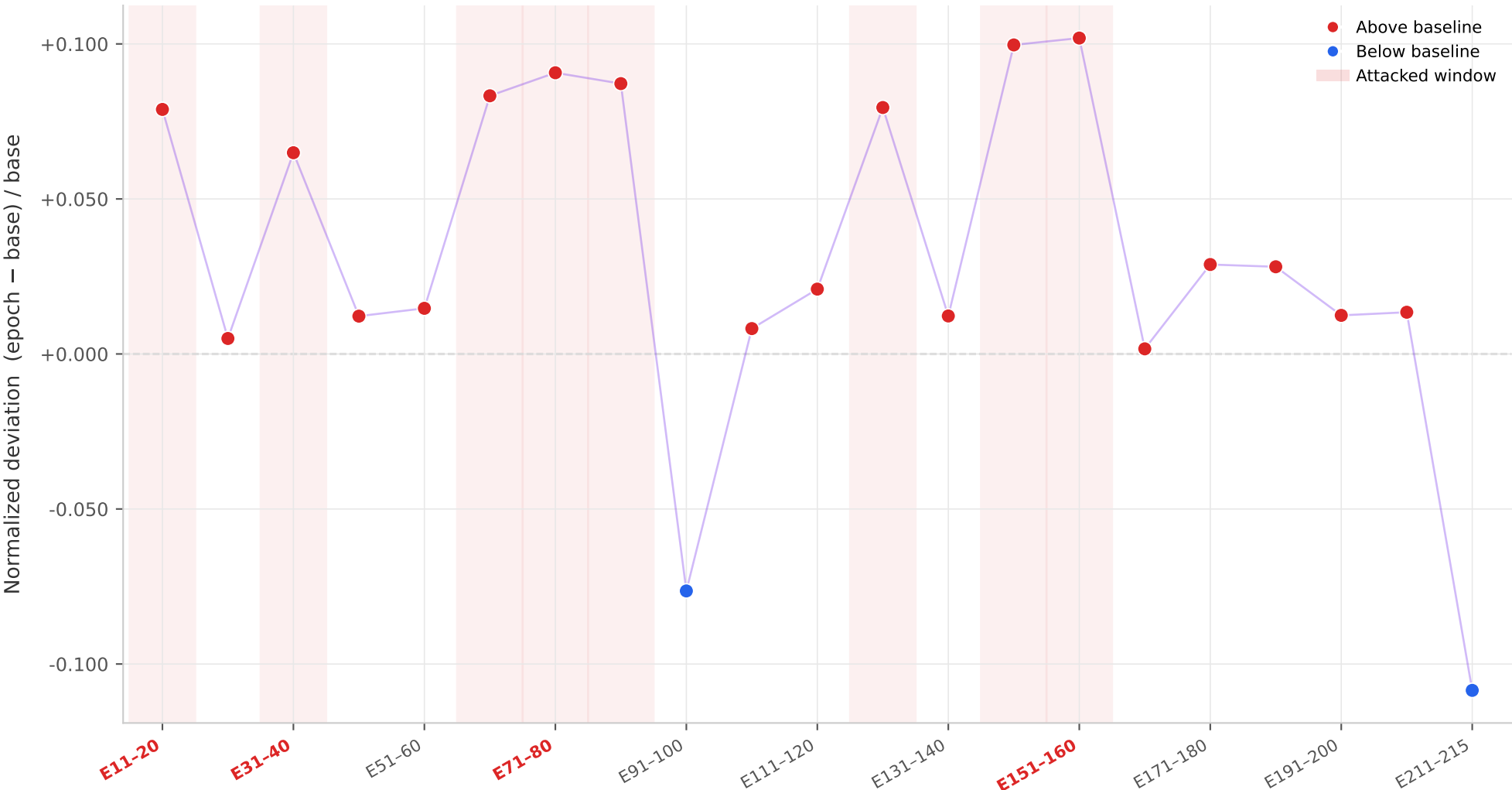
Flow



baseline = 5056 mean norm = +0.0172 max |norm| = 0.1156 malicious = 5.0% baseline epochs = 5.0%

Unique Source IPs

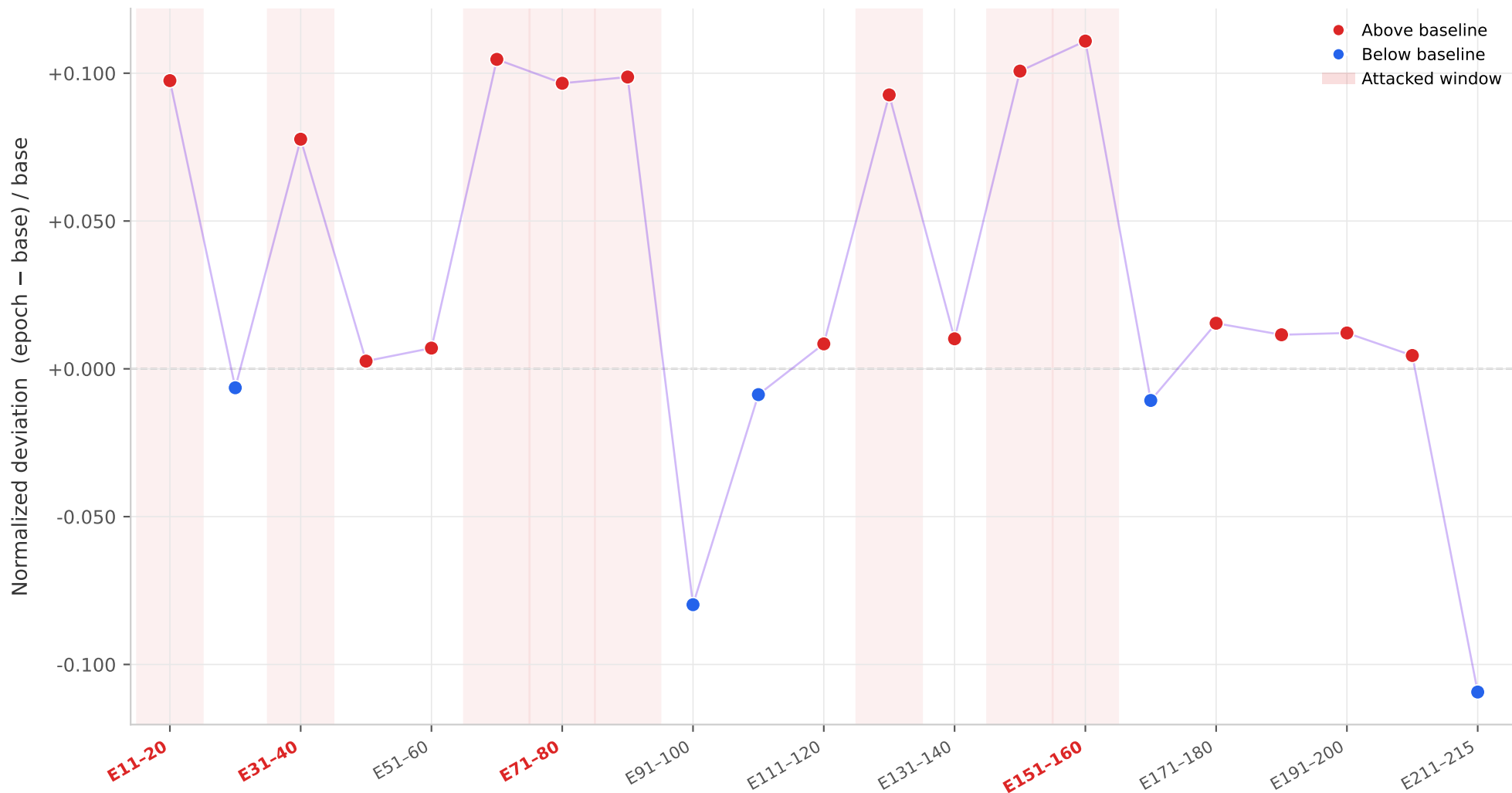
Count



baseline = 3567 mean norm = +0.0314 max |norm| = 0.1085 malicious = 5.0% baseline epochs = 5.0%

Unique Destination IPs

Count

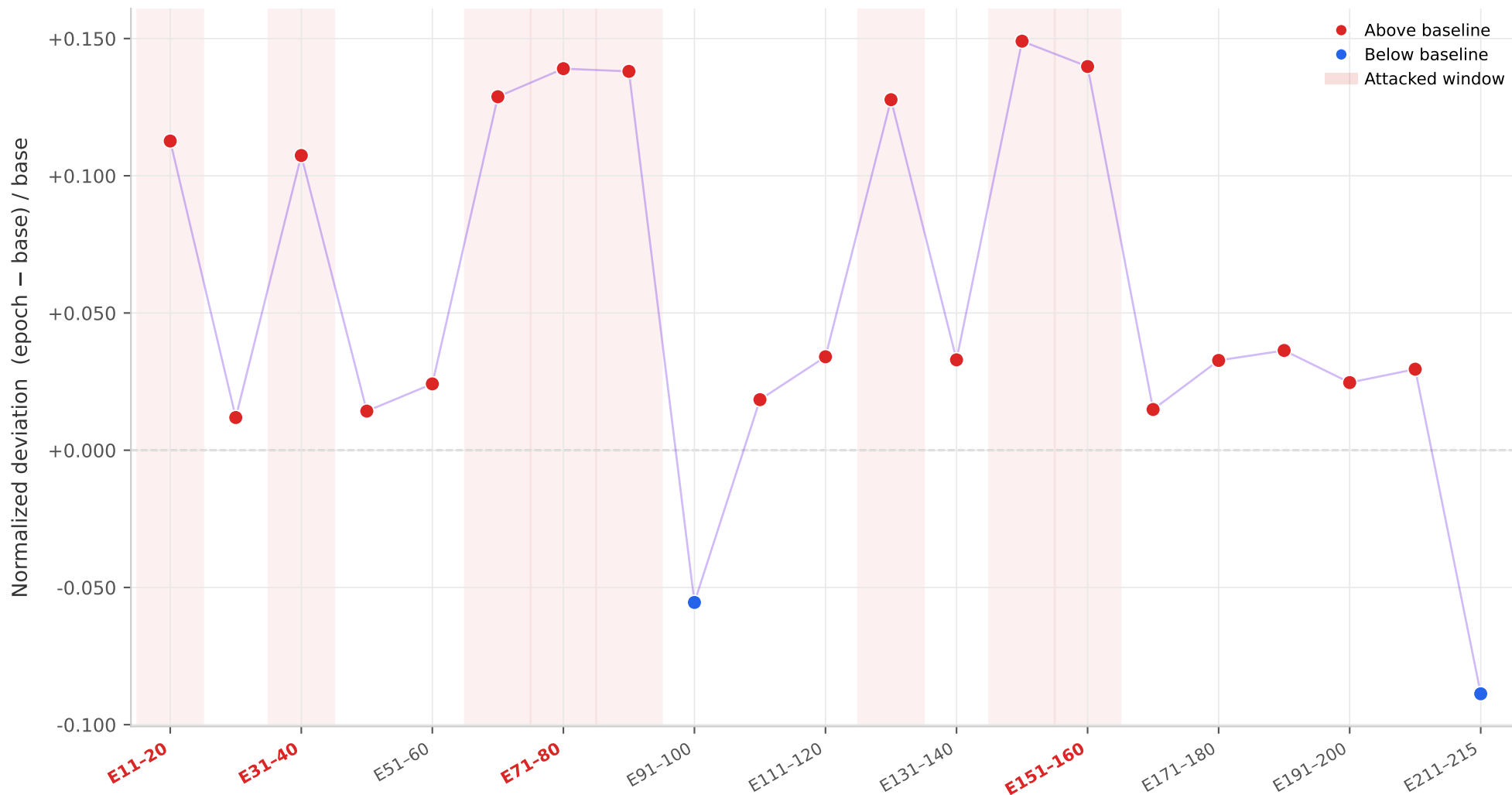


Epoch group (each point = 10 epochs)

baseline = 2795 mean norm = +0.0303 max |norm| = 0.1109 malicious = 5.0% baseline epochs = 5.0%

Unique Source Ports

Count



Epoch group (each point = 10 epochs)

baseline = 2317 mean norm = +0.0558 max |norm| = 0.1490 malicious = 5.0% baseline epochs = 5.0%

Unique Destination Ports

Count

